

Cover Page
SQL DATA ANALYSIS
Project 3
Industrial Training Kit
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Submitted To
DecodeLabs
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Introduction

SQL (Structured Query Language) is the standard language used to interact with relational databases. It allows users to retrieve, filter, sort, group, and analyze data efficiently. In this project, SQL was used to analyze an e-commerce sales dataset imported into MySQL. Various SQL queries were executed to generate business insights, including filtering records, sorting data, grouping information, and performing aggregate calculations such as COUNT(), SUM(), and AVG().

Objectives

- Learn the fundamentals of SQL.
- Retrieve data using SELECT statements.
- Filter records using the WHERE clause.
- Sort data using ORDER BY.
- Group records using GROUP BY.
- Perform aggregate calculations using COUNT(), SUM(), and AVG().
- Generate meaningful business insights from an e-commerce sales dataset.

Tools and Technologies Used

Tool	Purpose
MySQL Server 8.0	Database management
MySQL Workbench	SQL query execution
Microsoft Excel	Dataset preparation
Git	Version control
GitHub	Project hosting
Microsoft Word	Report preparation

Dataset Description

The dataset contains 1200 records and 14 columns representing e-commerce sales transactions. It includes customer information, product details, payment methods, order status, and sales values. The dataset was imported into MySQL for SQL analysis.

	Column	Description	
	OrderID	Unique order ID	
	Date	Order date	

CustomerID	Customer ID	
Product	Product name	
Quantity	Quantity purchased	
UnitPrice	Price per unit	
ShippingAddress	Delivery address	
PaymentMethod	Payment method	
OrderStatus	Status of the order	
TrackingNumber	Tracking ID	
ItemsInCart	Number of items in cart	
CouponCode	Coupon applied	
ReferralSource	Source of customer	
TotalPrice	Total order value	

Database Creation and Data Import

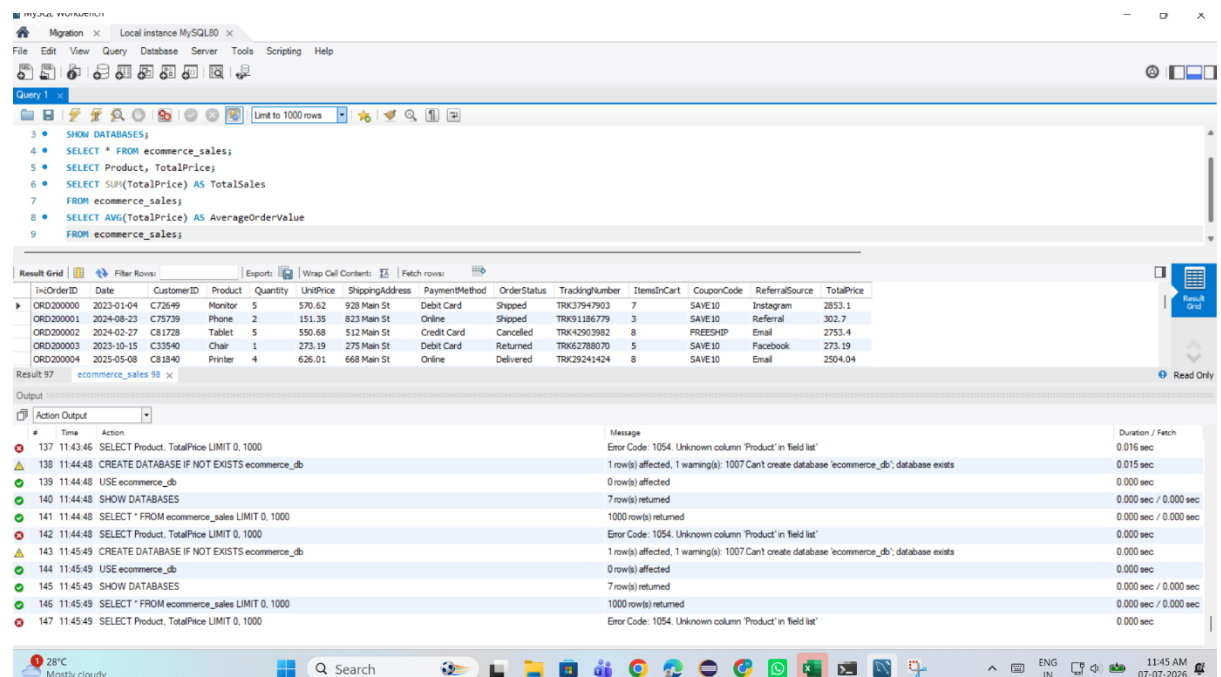
Database Creation

```
CREATE DATABASE ecommerce_db;
USE ecommerce_db;
```

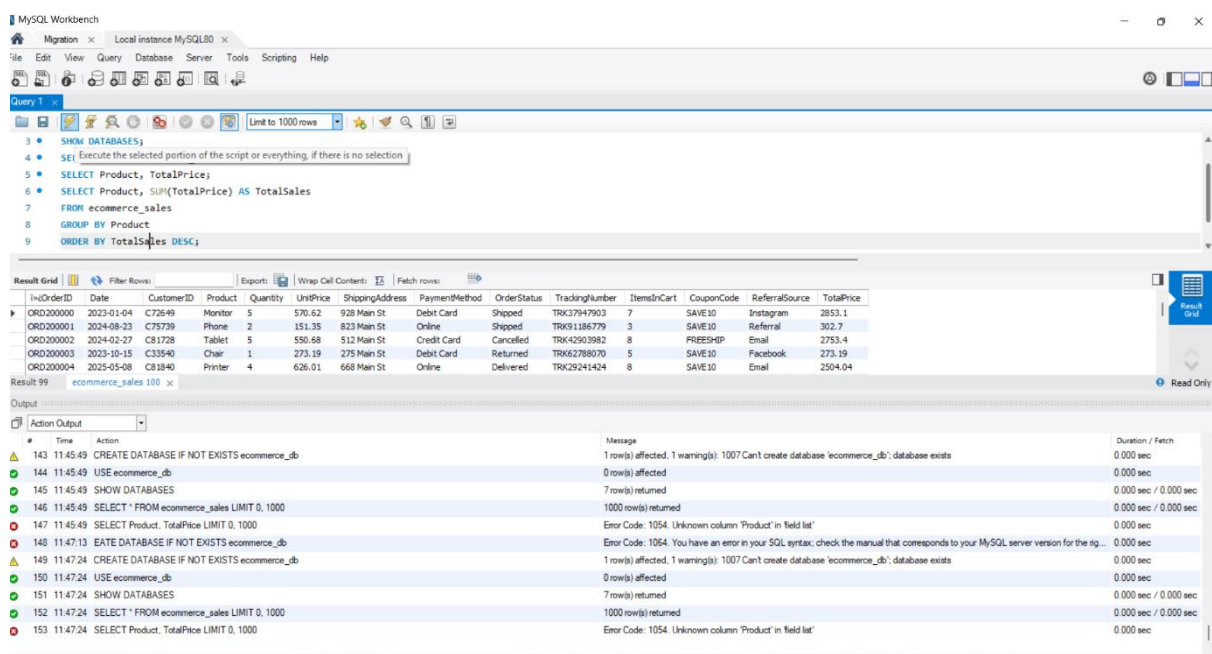
Data Import

The cleaned dataset was converted from Excel (.xlsx) to CSV (.csv) and imported into MySQL using the Table Data Import Wizard in MySQL Workbench. After the import, the data was stored in the ecommerce_sales table for analysis.

SQL coding Screenshots



(1)



(2)

SQL Queries and Results

6.1 Sort Records Using ORDER BY

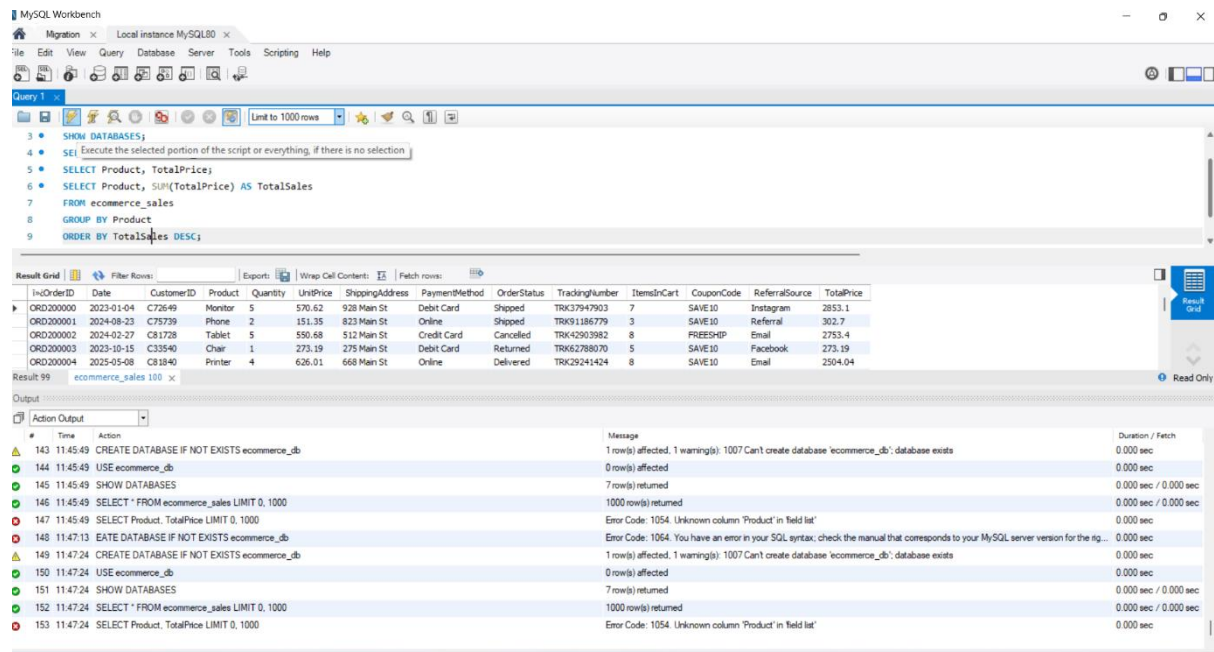
SELECT *

FROM ecommerce_sales

ORDER BY TotalPrice DESC;

Description

This query sorts all orders from the highest total price to the lowest



The screenshot shows the MySQL Workbench interface. The 'Query' tab is active, displaying a SQL query that sorts orders by total price in descending order. The 'Result Grid' shows the output of the query, listing various orders with their details. The 'Output' tab shows the execution log, indicating that the database 'ecommerce_db' was created and the query was executed successfully.

MySQL Workbench

Migration Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Query 1

```
3 SHOW DATABASES;
4 SE;
5 SELECT Product, TotalPrice;
6 SELECT Product, SUM(TotalPrice) AS TotalSales
7 FROM ecommerce_sales
8 GROUP BY Product
9 ORDER BY TotalSales DESC;
```

Result Grid

InOrderID	Date	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	CouponCode	ReferralSource	TotalPrice
ORD200000	2023-01-04	C72649	Monitor	5	570.62	928 Main St	Debit Card	Shipped	TRK37947903	7	SAVE10	Instagram	2853.1
ORD200001	2024-08-23	C75739	Phone	2	151.35	823 Main St	Online	Shipped	TRK91186779	3	SAVE10	Referral	302.7
ORD200002	2024-02-27	C81728	Tablet	5	550.68	512 Main St	Credit Card	Cancelled	TRK42903982	8	FREESHIP	Email	2753.4
ORD200003	2023-10-15	C33540	Chair	1	273.19	275 Main St	Debit Card	Returned	TRK62788070	5	SAVE10	Facebook	273.19
ORD200004	2025-05-08	C81840	Printer	4	626.01	668 Main St	Online	Delivered	TRK29241424	8	SAVE10	Email	2504.04

Result 99 ecommerce_sales 100

Output

#	Time	Action	Message	Duration / Fetch
143	11:45:49	CREATE DATABASE IF NOT EXISTS ecommerce_db	1 row(s) affected, 1 warning(s): 1007 Can't create database 'ecommerce_db': database exists	0.000 sec
144	11:45:49	USE ecommerce_db	0 row(s) affected	0.000 sec
145	11:45:49	SHOW DATABASES	7 row(s) returned	0.000 sec / 0.000 sec
146	11:45:49	SELECT * FROM ecommerce_sales LIMIT 0, 1000	1000 row(s) returned	0.000 sec / 0.000 sec
147	11:45:49	SELECT Product, TotalPrice LIMIT 0, 1000	Error Code: 1054. Unknown column 'Product' in field list	0.000 sec
148	11:47:13	EATE DATABASE IF NOT EXISTS ecommerce_db	Error Code: 1054. You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the rig...	0.000 sec
149	11:47:24	CREATE DATABASE IF NOT EXISTS ecommerce_db	1 row(s) affected, 1 warning(s): 1007 Can't create database 'ecommerce_db': database exists	0.000 sec
150	11:47:24	USE ecommerce_db	0 row(s) affected	0.000 sec
151	11:47:24	SHOW DATABASES	7 row(s) returned	0.000 sec / 0.000 sec
152	11:47:24	SELECT * FROM ecommerce_sales LIMIT 0, 1000	1000 row(s) returned	0.000 sec / 0.000 sec
153	11:47:24	SELECT Product, TotalPrice LIMIT 0, 1000	Error Code: 1054. Unknown column 'Product' in field list	0.000 sec

6.2 Group Records Using GROUP BY

SELECT Product,

COUNT(*) AS TotalOrders

FROM ecommerce_sales

GROUP BY Product;

Description

This query counts the number of orders for each product.

MySQL Workbench interface showing a query execution. The query is:

```

2 • USE ecommerce_db;
3 • SHOW DATABASES;
4 • SELECT * FROM ecommerce_sales;
5 • SELECT Product, TotalPrice;
6 • SELECT Product, COUNT(*) AS TotalOrders
7   FROM ecommerce_sales
8  GROUP BY Product;

```

The Result Grid shows 93 rows of data. The Output pane shows the execution log:

#	Time	Action	Message	Duration / Fetch
127	11:41:42	CREATE DATABASE IF NOT EXISTS ecommerce_db	1 row(s) affected, 1 warning(s): 1007 Can't create database 'ecommerce_db': database exists	0.000 sec
128	11:41:42	USE ecommerce_db	0 row(s) affected	0.000 sec
129	11:41:42	SHOW DATABASES	7 row(s) returned	0.000 sec / 0.000 sec
130	11:41:42	SELECT * FROM ecommerce_sales LIMIT 0, 1000	1000 row(s) returned	0.000 sec / 0.000 sec
131	11:41:42	SELECT Product, TotalPrice FROM ecommerce_sales LIMIT 0, 1000	1000 row(s) returned	0.000 sec / 0.000 sec
132	11:41:42	SELECT * FROM ecommerce_sales WHERE TotalPrice > 1000 LIMIT 0, 1000	512 row(s) returned	0.000 sec / 0.000 sec
133	11:43:46	CREATE DATABASE IF NOT EXISTS ecommerce_db	1 row(s) affected, 1 warning(s): 1007 Can't create database 'ecommerce_db': database exists	0.016 sec
134	11:43:46	USE ecommerce_db	0 row(s) affected	0.000 sec
135	11:43:46	SHOW DATABASES	7 row(s) returned	0.000 sec / 0.000 sec
136	11:43:46	SELECT * FROM ecommerce_sales LIMIT 0, 1000	1000 row(s) returned	0.000 sec / 0.000 sec
137	11:43:46	SELECT Product, TotalPrice LIMIT 0, 1000	Error Code: 1054. Unknown column 'Product' in field list	0.016 sec

6.3 SUM()

Query

SELECT SUM(TotalPrice) AS TotalSales

FROM ecommerce_sales;

Description

This query calculates the total sales amount from all orders.

MySQL Workbench interface showing a query execution. The query is:

```

3 • SHOW DATABASES;
4 • SELECT * FROM ecommerce_sales;
5 • SELECT Product, TotalPrice;
6 • SELECT SUM(TotalPrice) AS TotalSales
7   FROM ecommerce_sales;
8
9

```

The Result Grid shows 95 rows of data. The Output pane shows the execution log:

#	Time	Action	Message	Duration / Fetch
132	11:41:42	SELECT * FROM ecommerce_sales WHERE TotalPrice > 1000 LIMIT 0, 1000	512 row(s) returned	0.000 sec / 0.000 sec
133	11:43:46	CREATE DATABASE IF NOT EXISTS ecommerce_db	1 row(s) affected, 1 warning(s): 1007 Can't create database 'ecommerce_db': database exists	0.016 sec
134	11:43:46	USE ecommerce_db	0 row(s) affected	0.000 sec
135	11:43:46	SHOW DATABASES	7 row(s) returned	0.000 sec / 0.000 sec
136	11:43:46	SELECT * FROM ecommerce_sales LIMIT 0, 1000	1000 row(s) returned	0.000 sec / 0.000 sec
137	11:43:46	SELECT Product, TotalPrice LIMIT 0, 1000	Error Code: 1054. Unknown column 'Product' in field list	0.016 sec
138	11:44:48	CREATE DATABASE IF NOT EXISTS ecommerce_db	1 row(s) affected, 1 warning(s): 1007 Can't create database 'ecommerce_db': database exists	0.015 sec
139	11:44:48	USE ecommerce_db	0 row(s) affected	0.000 sec
140	11:44:48	SHOW DATABASES	7 row(s) returned	0.000 sec / 0.000 sec
141	11:44:48	SELECT * FROM ecommerce_sales LIMIT 0, 1000	1000 row(s) returned	0.000 sec / 0.000 sec
142	11:44:48	SELECT Product, TotalPrice LIMIT 0, 1000	Error Code: 1054. Unknown column 'Product' in field list	0.000 sec

6.4 AVG()

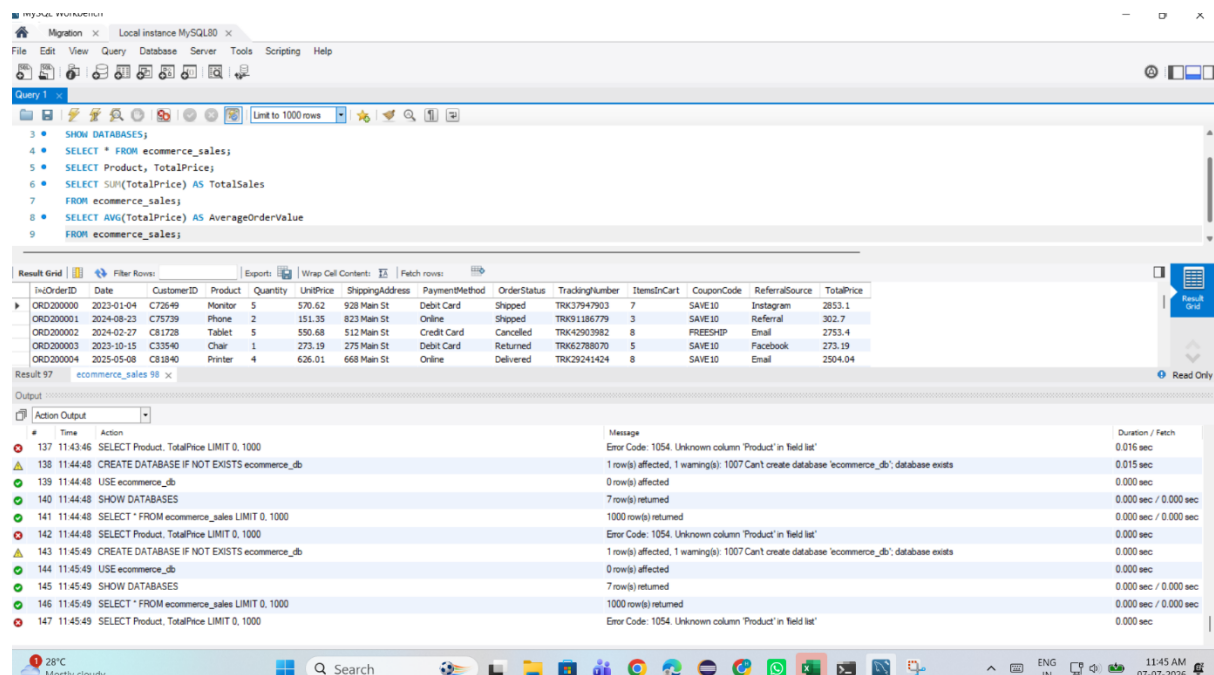
Query

```
SELECT AVG(TotalPrice) AS AverageOrderValue
```

```
FROM ecommerce_sales;
```

Description

This query calculates the average value of all customer orders.



The screenshot shows the MySQL Workbench interface. The 'Query' tab is active, displaying a SQL query that calculates the average order value. The 'Result Grid' tab shows the results of the query, which includes a table with columns: InvOrderID, Date, CustomerID, Product, Quantity, UnitPrice, ShippingAddress, PaymentMethod, OrderStatus, TrackingNumber, ItemsInCart, CouponCode, ReferralSource, and TotalPrice. The results show several orders, including Monitor, Phone, Tablet, Chair, and Printer. The 'Output' tab shows the execution log, indicating that the query was successful and returned 1000 rows.

InvOrderID	Date	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	CouponCode	ReferralSource	TotalPrice
ORD200000	2023-01-04	C72649	Monitor	5	570.62	928 Main St	Debit Card	Shipped	TRK37947903	7	SAVE10	Instagram	2853.1
ORD200001	2024-08-23	C75739	Phone	2	151.35	823 Main St	Online	Shipped	TRK91186779	3	SAVE10	Referral	302.7
ORD200002	2024-02-27	C81728	Tablet	5	550.68	512 Main St	Credit Card	Cancelled	TRK42903982	8	FREESHIP	Email	2753.4
ORD200003	2023-10-15	C33540	Chair	1	273.19	275 Main St	Debit Card	Returned	TRK62788070	5	SAVE10	Facebook	273.19
ORD200004	2025-05-08	C81840	Printer	4	626.01	668 Main St	Online	Delivered	TRK29241424	8	SAVE10	Email	2504.04

Key Findings

- The dataset contains **1200 e-commerce transactions**.
- SQL was successfully used to retrieve, filter, sort, and group data.
- Aggregate functions (COUNT, SUM, and AVG) provided meaningful business metrics.
- Product-wise sales analysis identified the distribution of orders across products.
- Payment method analysis showed customer purchasing preferences.
- Referral source analysis highlighted the channels bringing customers to the platform.
- SQL enables efficient extraction of insights from relational databases.

Conclusion

This project demonstrated the practical use of SQL for analyzing an e-commerce sales dataset. By applying SQL concepts such as SELECT, WHERE, ORDER BY, GROUP BY, COUNT(), SUM(), and AVG(), meaningful business insights were generated from raw transactional data. The project strengthened my understanding of database querying and

data analysis, while also improving my ability to organize and interpret information stored in relational databases.

References

- [MySQL Documentation](#)
- [MySQL Workbench](#)
- [Microsoft Excel Documentation](#)
- [DecodeLabs Industrial Training Kit](#)